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Thermo-mechanical verification of a fuel pin of an AP1000 reactor

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Chapter 1

Introduction

Fuel rods inside a nuclear reactor are subjected to intense thermo-mechanical stresses during operations, primarily driven by pressure loads and radial thermal gradients. This document outlines the thermo-mechanical verification of a fuel pin cladding based on the AP1000 reactor design parameters.

The work is focused on achieving two main objectives:

- **Structural verification:** calculate the mechanical and thermal stresses (in their hoop/tangential, radial, and longitudinal components) and demonstrate that the equivalent stresses comply with the safety limits prescribed by the ASME BPVC Section III standard
- **Plenum sizing:** estimate the production of fission gases and geometrically size the gas plenum (volume and height) to ensure the internal pressure does not exceed the limit values established during the design phase.

1.1 Main assumptions and simplifications

The following simplifying assumptions have been made for the formulation and resolution of this problem:

- The reactor is assumed to be operating at steady-state under nominal conditions
- Only static loads are considered.
- Bending stresses, peak stresses, and local stresses are considered negligible and are therefore excluded from the analysis.
- The cladding material is assumed to be Zircaloy-4 instead of ZIRLO.
- The maximum internal pressure (used to evaluate mechanical stresses under design conditions) is calculated assuming a conservative limit scenario of reactor depressurization ($p_o \sim 0$).
- All gases inside the fuel pin, including fission gases (Xe and Kr) and impurities (N_2 and H_2O vapor), are treated as ideal gases.
- The available volume for gas expansion is assumed to be exclusively that of the gas plenum; the gap between the fuel pellet and the cladding is not considered as available volume.
- The fuel enrichment (ϵ) is assumed to be 3%.

1.2 Main data

Component	Parameter	Symbol	Value	Unit
Power & Heat Flux	Reactor type	-	PWR	-
	Coolant	-	Light water	-
	Moderator	-	Light water	-
	Reactor core heat output	P_{th}	3400	MW _t
	Heat generated in fuel	-	97.4	%
	Average heat flux	q''_{avg}	0.6287	MW/m ²
	Maximum heat flux (normal op.)	q''_{max}	1634709	W/m ²
	Average linear power	q'_{avg}	18766	W/m
	Peak linear power (normal op.)	q'_{peak}	48884	W/m
	Power peaking factor	F_Q	2.60	-
	Power density	q'''	109.7×10^6	W/m ³
	Gravity acceleration	g	9.81	m/s ²
Pressure	System pressure, nominal	P_{nom}	155.13	bar
	System pressure, min steady-state	P_{min}	150.99	bar
	Pressure drop across core	ΔP_{core}	2.75	bar
	Pressure drop across vessel	ΔP_{vessel}	4.30	bar
Coolant Flow	Total flow	$\dot{m}_{tot}$	14300	kg/s
	Effective flow	$\dot{m}_{eff}$	13460	kg/s
	Flow area	A_{eff}	3.883	m ²
	Velocity	v	4.82	m/s
	Mass velocity	G	3460	kg/(s·m ²)
Temperatures	Inlet temperature	T_{in}	279.4	°C
	Rise vessel	ΔT_v	42.9	°C
	Rise core	ΔT_c	45.2	°C
	Average core temperature	$T_{avg,c}$	303.4	°C
	Average vessel temperature	$T_{avg,v}$	300.89	°C
Heat Transfer	Heat transfer area	A_{ht}	5268	m ²
	Max heat flux	q''_{max}	1.635×10^6	W/m ²
	Peak linear power	q'_{peak}	48.88	kW/m
	Max peak linear power	q'_{max}	73.66	kW/m

Table 1.1: Thermal and Hydraulic Design Parameters

Component	Parameter	Symbol	Value	Unit
Core Geometry	Number of fuel assemblies	N_{ass}	157	-
	Core diameter	D_{core}	3.040	m
	Core height	H	4.2672	m
Fuel Assembly	Fuel assembly type	-	17×17 XL	-
	Rods per assembly	$N_{rods/ass}$	264	-
	Rod pitch	w	0.01260	m
	Assembly width	L_{ass}	0.21402	m
	Fuel weight	W_{fuel}	95974.7	kg
	Clad weight	W_{clad}	19552	kg
	Top/Bottom Grids	-	2	-
	Intermediate Grids	-	8	-
	IFM Grids	-	4	-
	Protective Grids	-	1	-
Fuel Rods	Total rods	N_{rods}	41448	-
	Rod diameter	D_{ro}	0.00950	m
	Gap	t_{gap}	0.000165	m
	Clad thickness	t_c	0.0005715	m
Fuel Pellets	Material	-	UO ₂	-
	Density (% of theoretical)	ρ	95.5	%
	Diameter	D_{pel}	0.00819	m
	Length	h_{pel}	0.00983	m
Control Rods	RCCA clusters	-	53	-
	GRCA clusters	-	16	-
	Absorber material	-	Ag-In-Cd	-
	Rodlets per cluster	-	24	-
Safety Limits	Max clad temperature	$T_{c,max}$	1204	°C
	Max fuel temp (melt limit)	$T_{f,max}$	2593.33	°C
	Max control speed	v_{cr}	1.143	m/min
	Min DNBR (typical channel)	-	2.80	-
	Min DNBR (thimble cold wall)	-	2.74	-
	Core avg void fraction	-	0.0	%
	Hot subchannel avg void frac.	-	0.1	%
Strength	Hot subchannel max void frac.	-	0.9	%
	Yield strength	σ_y	241	MPa
	Ultimate tensile strength	σ_u	413	MPa

Table 1.2: Core Design, Fuel Assemblies, Control Rods and Safety Limits

Chapter 2

Calculations

2.1 Preliminary verification against buckling

The first structural assessment involves verifying the cladding's resistance to elastic instability, commonly known as buckling. During the early stages of reactor operation, the internal gas pressure is minimal, and the fuel pin is subjected to a severe external compressive load driven by the primary coolant system pressure ($P = 155.13$ bar). To ensure the cladding does not collapse, the critical buckling pressure p_{cr} is calculated and compared against the nominal system pressure. The critical pressure is evaluated using the classical formulation for cylindrical shells

$$p_{cr} = \frac{E}{4(1 - \nu^2)} \left(\frac{t}{r_{avg}} \right)^3$$

where:

- E is the Young's modulus of the Zircaloy-4 cladding
- ν is the Poisson's ratio of the material
- t is the cladding thickness
- r_{avg} is the average distance of the cladding from the fuel pellet centerline.

The thermo-mechanical properties of Zircaloy-4 (E and ν) are highly dependent on the local operating temperature. To perform a rigorous and conservative engineering verification, the material properties were evaluated at the maximum temperature node along the cladding's axial development. At this thermal peak, the elastic modulus experiences its most significant reduction, representing the most structurally vulnerable condition for the component's stability.

Based on the specified AP1000 geometry, the average cladding radius evaluates to $r_{avg} = 4.4642$ mm. The analytical evaluation yields a minimum critical buckling pressure of $\mathbf{p_{cr} = 459.70 \text{ bar}}$. Comparing this capacity against the external nominal system pressure of $P = 155.13$ bar, the cladding demonstrates a substantial safety margin of 304.57 bar.

Since the stability condition $\mathbf{p_{cr} > P}$ is widely satisfied even under the most unfavorable thermal conditions, the structural integrity of the AP1000 fuel pin cladding is fully verified against elastic buckling under steady-state nominal operations.

2.2 Maximum internal pressure in the cladding

To perform a robust thermo-mechanical verification of the cladding, it is necessary to determine the maximum internal pressure that the fuel pin can withstand. Following the assignment guidelines, this bounding calculation assumes a conservative depressurized reactor scenario where the external system pressure is completely lost ($p_o = 0$).

As a first approximation, the maximum allowable internal pressure is reached when the peak tangential (hoop) stress, σ_h , equals the yield strength of the Zircaloy-4 cladding ($\sigma_y = 241$ MPa).

To determine the appropriate analytical model, the structural geometry must be evaluated. The threshold for classifying a cylinder as thin-walled is given by the geometric ratio:

$$\frac{t}{r_{\text{avg}}} \ll 1 \quad (2.1)$$

Since $\ll$ can be interpreted at least as $<$ of a tenth of a quantity, in this specific case:

$$\frac{t}{r_{\text{avg}}} = 0.13 > 0.1 \quad (2.2)$$

Since this ratio exceeds the threshold, the cladding cannot be considered as a thin shell; it must be rigorously classified and treated as a **thick-walled cylinder**. Consequently, simple membrane formulas are no longer valid, and **Lamé's equations** must be applied to account for the non-uniform stress distribution across the cladding thickness.

The maximum stress intensity in a thick-walled cylinder subjected to internal pressure occurs at the innermost radius ($r = r_i$). Under the depressurized assumption ($p_o = 0$), principal stress components at the inner wall given by Lamé's equations are:

$$\sigma_h = p_{i,\text{max}} \frac{r_o^2 + r_i^2}{r_o^2 - r_i^2} \quad (2.3)$$

$$\sigma_r = -p_{i,\text{max}} \quad (2.4)$$

$$\sigma_l = p_{i,\text{max}} \frac{r_i^2}{r_o^2 - r_i^2} \quad (2.5)$$

Following the ASME BPVC Section III approach, the equivalent stress is evaluated using Tresca's maximum shear stress criterion. At the inner wall, the dominant stress intensity

$$\sigma_h - \sigma_r = p_{i,\text{max}} \frac{r_o^2 + r_i^2}{r_o^2 - r_i^2} + p_{i,\text{max}} = p_{i,\text{max}} \frac{r_o^2 + r_i^2 + r_o^2 - r_i^2}{r_o^2 - r_i^2} = p_{i,\text{max}} \frac{2r_o^2}{r_o^2 - r_i^2} \quad (2.6)$$

Setting this equal to the yield strength σ_y (onset of plasticity according to Tresca):

$$p_{i,\text{max}} = \sigma_y \frac{r_o^2 - r_i^2}{2r_o^2} \quad (2.7)$$

Substituting the geometric and material properties, the maximum internal pressure is evaluated as:

$$p_{i,\text{max}} = 27.25 \text{ MPa} \quad (272.52 \text{ bar}) \quad (2.8)$$

2.2.1 Results and analysis

Given the Zircaloy-4 yield strength of 241 MPa and the AP1000 cladding geometry evaluated in the previous steps, the rigorous thick-wall formulation yields a maximum internal pressure of 272.52 bar (27.25 MPa). Under this bounding pressure limit, the three principal stress components acting on the inner wall of the cladding are as:

- **Tangential (hoop) stress (σ_h):** 241 MPa. Tresca is the combination of this stress and the radial stress (the maximum stress intensity, $\sigma_h - \sigma_r$) that acts as the limiting parameter, set equal to the material's yield strength (σ_y) to define the plastic deformation at the inner radius.
- **Longitudinal stress (σ_l):** 93.25 MPa. This represents the axial tension induced by the internal pressure on a finite cylinder.

- **Radial stress (σ_r):** -27.25 MPa. This is equal to the negative of the maximum internal pressure ($-p_{i,\max}$), representing the pure compressive boundary load acting directly on the innermost surface of the cladding.

This calculated pressure of 272.52 bar serves as a highly conservative upper bound. It represents the absolute maximum internal gas accumulation the cladding can withstand before yielding begins under the assumption of a total loss of external counter-pressure. These principal stress values, obtained at the most severely loaded location (the inner radius), will serve as the primary inputs for the subsequent ASME BPVC Section III structural integrity and stress intensity verifications.

2.3 Mechanical stresses

Once the maximum allowable internal pressure $p_{i,\max}$ has been determined, the comprehensive stress field across the cladding wall must be mapped. To account for the thick-walled geometry demonstrated by the t/r_{avg} ratio, the analysis consistently relies on **Lamé's elastic equations** for thick cylinders.

Under the specific bounding conditions of internal pressure $p_i = p_{i,\max}$ and external refrigerant pressure under operating condition ($p_o = 15.513$ MPa), Lamé's general formulations for tangential (hoop), radial, and longitudinal stresses as a function of the generic radius r are expressed as follows:

- **At the inner cladding wall ($r = r_i$):** The radial stress matches the internal boundary condition (compressive stress equal to the internal pressure), while the hoop stress reaches its absolute maximum value across the thickness:

$$\sigma_{h,\text{mec}}^{\text{in}} = \frac{p_{i,\max}(r_o^2 + r_i^2) - 2p_o r_o^2}{r_o^2 - r_i^2} \quad (2.9)$$

$$\sigma_{r,\text{mec}}^{\text{in}} = -p_{i,\max} \quad (2.10)$$

$$\sigma_{l,\text{mec}}^{\text{in}} = \frac{p_{i,\max} r_i^2 - p_o r_o^2}{r_o^2 - r_i^2} \quad (2.11)$$

- **At the outer cladding wall ($r = r_o$):** The radial stress matches the external operating pressure boundary condition, while the hoop stress updates according to the external boundary geometry.

$$\sigma_{h,\text{mec}}^{\text{out}} = \frac{2p_{i,\max} r_i^2 - p_o(r_o^2 + r_i^2)}{r_o^2 - r_i^2} \quad (2.12)$$

$$\sigma_{r,\text{mec}}^{\text{out}} = -p_o = -15.51\text{MPa} \quad (2.13)$$

$$\sigma_{l,\text{mec}}^{\text{out}} = \frac{p_{i,\max} r_i^2 - p_o r_o^2}{r_o^2 - r_i^2} \quad (2.14)$$

The mechanical stresses values are shown in the Table 3:

Table 2.1: Summary table with mechanical stresses considering thick walls (Lamé's Equations).

Component	Inner cladding wall [MPa]	Outer cladding wall [MPa]
Hoop (tangential)	76.56	64.82
Radial	-27.25	-15.51
Longitudinal	24.65	24.65

2.4 Thermal stresses

Thermal stresses in nuclear fuel cladding arise due to the constrained thermal expansion resulting from the severe temperature gradient across the cladding wall. During reactor operation, heat generated by nuclear fission in the UO_2 pellet is transferred radially outwards through the cladding to the coolant. This creates a non-uniform temperature profile, where the inner cladding surface temperature (T_{ci}) is significantly higher than the outer surface temperature (T_{co}).

Because the hotter inner fibers attempt to expand more than the cooler outer fibers, they are placed under compressive stress. Conversely, the cooler outer fibers restrict this expansion and are subjected to tensile stress.

Assuming a steady-state radial temperature distribution governed by Fourier's law of heat conduction, the temperature profile $T(r)$ as a function of the radius r inside a thick-walled cylinder is given by:

$$T(r) = T_{ci} - \Delta T_{cl} \frac{\ln(r/r_i)}{\ln(r_o/r_i)} \quad (2.15)$$

where $\Delta T_{cl} = T_{ci} - T_{co}$ represents the temperature drop across the cladding wall thickness.

By applying the thermoelastic stress equations for an unconstrained axisymmetrical cylinder under a radial temperature gradient, the principal thermal stress components (tangential/hoop ($\sigma_{h,th}$), longitudinal ($\sigma_{l,th}$), and radial ($\sigma_{r,th}$)) can be analytically determined.

At the boundaries, the radial thermal stress is zero ($\sigma_{r,th}(r_i) = \sigma_{r,th}(r_o) = 0$).

In order to do it, the analytical thermal factors for the inner and outer walls are defined as:

$$\Psi_{in} = \frac{\alpha_{in} E_{in} \Delta T_{cl,max}}{2(1 - \nu_{in}) \ln(r_o/r_i)} \quad (2.16)$$

$$\Psi_{out} = \frac{\alpha_{out} E_{out} \Delta T_{cl,max}}{2(1 - \nu_{out}) \ln(r_o/r_i)} \quad (2.17)$$

Where:

- E is the temperature-dependent Young's Modulus of Zircaloy-4.
- ν is the Poisson's ratio.
- α is the linear thermal expansion coefficient.
- r_i and r_o are the inner and outer cladding radii, respectively.

Consequently, the thermal stresses are evaluated:

- **At the inner cladding wall ($r = r_i$):**

The hoop and longitudinal thermal stresses are identical and compressive, while the radial thermal stress is null.

$$\sigma_{h,th}^{in} = \sigma_{l,th}^{in} = -\Psi_{in} \left[\left(\frac{2r_o^2}{r_o^2 - r_i^2} \right) \ln \left(\frac{r_o}{r_i} \right) - 1 \right] \quad (2.18)$$

$$\sigma_{r,th}^{in} = 0 \quad (2.19)$$

- **At the outer cladding wall ($r = r_o$):**

the thermal stresses are tensile, while the radial one is null:

$$\sigma_{h,th}^{out} = \sigma_{l,th}^{out} = \Psi_{out} \left[1 - \left(\frac{2r_i^2}{r_o^2 - r_i^2} \right) \ln \left(\frac{r_o}{r_i} \right) \right] \quad (2.20)$$

$$\sigma_{r,th}^{out} = 0 \quad (2.21)$$

The evaluated thermal stresses are shown in the Table 5.

Table 2.2: Thermal stress components evaluated at the peak temperature location across the AP1000 cladding wall.

Component	Inner Cladding Wall [MPa]	Outer Cladding Wall [MPa]
Hoop (tangential)	-24.26	23.07
Radial	0.00	0.00
Longitudinal	-24.26	23.07

2.5 Stress classification (ASME approach)

2.5.1 Primary stresses

Primary stresses are those developed by the imposed mechanical loads and are necessary to satisfy the simple laws of equilibrium of forces and moments. In a general core component design, primary stresses are divided into three distinct sub-components:

- **Membrane stress:** The average stress across the thickness of the component.
- **Bending stress:** The component of stress that varies linearly across the thickness.
- **Local stress:** Highly localized stresses occurring at geometric discontinuities.

According to the design assumptions established for this assignment, both bending stresses and local stresses are considered negligible in magnitude and are therefore neglected ($\sigma_b \approx 0$, $\sigma_{local} \approx 0$).

Consequently, the primary stress tensor is purely represented by the primary membrane stress, which is mathematically evaluated by taking the arithmetic mean between the inner and outer cladding wall values for each principal stress component ($\sigma_h, \sigma_r, \sigma_l$):

$$\sigma_{h,mec}^{avg} = \frac{\sigma_{h,mec}^{in} + \sigma_{h,mec}^{out}}{2} \quad (2.22)$$

$$\sigma_{r,mec}^{avg} = \frac{\sigma_{r,mec}^{in} + \sigma_{r,mec}^{out}}{2} \quad (2.23)$$

$$\sigma_{l,mec}^{avg} = \frac{\sigma_{l,mec}^{in} + \sigma_{l,mec}^{out}}{2} \quad (2.24)$$

Calculated primary stresses are reported in the Table 4.

Table 2.3: Summary table with average mechanical stresses (primary stresses).

Component	Average Primary Stress [MPa]
Average tangential (hoop) stress	70.69
Average radial stress	-21.38
Average longitudinal stress	24.65

2.5.2 Secondary stresses

In structural mechanics and nuclear design criteria, secondary stresses are defined as stresses developed by the self-constraint of a structure or by constraints from adjacent components. Unlike primary stresses, which are caused by external mechanical loads (like internal pressure) and are necessary to satisfy laws of equilibrium, secondary stresses are self-limiting. This means that localized plastic deformation or minor distortions can shed or relax the thermal and structural strain, preventing catastrophic failure from a single application of the load.

In a thick-walled cladding analysis, these stresses are composed of both a mechanical component (representing the localized deviation from the average primary membrane stress) and a thermal component. The principal secondary stress components (tangential/ loop ($\sigma_{h,sec}$), the radial ($\sigma_{r,sec}$) and the longitudinal ($\sigma_{l,sec}$)) are evaluated at both the inner and outer cladding boundaries.

- **At the Inner Cladding Wall ($r = r_i$):** The algebraic summation of the mechanical and thermal stress components on the inner boundary yields:

$$\sigma_{h,sec}^{in} = \left(\sigma_{h,mec}^{in} - \sigma_{h,mec}^{avg} \right) + \sigma_{h,th}^{in} \quad (2.25)$$

$$\sigma_{r,sec}^{in} = \left(\sigma_{r,mec}^{in} - \sigma_{r,mec}^{avg} \right) + \sigma_{r,th}^{in} \quad (2.26)$$

$$\sigma_{l,sec}^{in} = \left(\sigma_{l,mec}^{in} - \sigma_{l,mec}^{avg} \right) + \sigma_{l,th}^{in} \quad (2.27)$$

- **At the Outer Cladding Wall ($r = r_o$):**

Similarly, the secondary stress state on the outer fiber is calculated as:

$$\sigma_{h,sec}^{out} = \left(\sigma_{h,mec}^{out} - \sigma_{h,mec}^{avg} \right) + \sigma_{h,th}^{out} \quad (2.28)$$

$$\sigma_{r,sec}^{out} = \left(\sigma_{r,mec}^{out} - \sigma_{r,mec}^{avg} \right) + \sigma_{r,th}^{out} \quad (2.29)$$

$$\sigma_{l,sec}^{out} = \left(\sigma_{l,mec}^{out} - \sigma_{l,mec}^{avg} \right) + \sigma_{l,th}^{out} \quad (2.30)$$

The evaluated secondary stresses are shown in the Table 6.

Table 2.4: Secondary stress components (combining mechanical and thermal components) evaluated at the inner and outer cladding walls in the peak points.

Component	Inner Cladding Wall [MPa]	Outer Cladding Wall [MPa]
Hoop (tangential)	-18.39	17.20
Radial	-5.87	5.87
Longitudinal	-24.26	23.07

2.5.3 Primary + Secondary stresses

The total stress state within the fuel pin cladding represents the actual superposition of all acting structural loads under normal operating conditions. It is computed via the direct algebraic sum of the local mechanical stresses and the thermal stresses at each respective boundary.

As noted during the analytical breakdown, the average primary membrane mechanical stresses cancel out when summing primary and secondary components explicitly, leaving the direct combination of local mechanical and thermal contributions. These total principal stress components (hoop ($\sigma_{h,tot}$), radial ($\sigma_{r,tot}$), and longitudinal ($\sigma_{l,tot}$)) determine the peak stress intensity used to perform the structural integrity verification according to the ASME Code Section III.

- **At the Inner Cladding Wall ($r = r_i$):** The total stress components acting on the inner fiber of the cladding are defined as:

$$\sigma_{h,tot}^{in} = \sigma_{h,mec}^{in} + \sigma_{h,th}^{in} \quad (2.31)$$

$$\sigma_{r,tot}^{in} = \sigma_{r,mec}^{in} + \sigma_{r,th}^{in} \quad (2.32)$$

$$\sigma_{l,tot}^{in} = \sigma_{l,mec}^{in} + \sigma_{l,th}^{in} \quad (2.33)$$

- **At the Outer Cladding Wall ($r = r_o$):** Similarly, the total stress components acting on the outer fiber of the cladding are calculated as:

$$\sigma_{h,tot}^{out} = \sigma_{h,mec}^{out} + \sigma_{h,th}^{out} \quad (2.34)$$

$$\sigma_{r,tot}^{out} = \sigma_{r,mec}^{out} + \sigma_{r,th}^{out} \quad (2.35)$$

$$\sigma_{l,tot}^{out} = \sigma_{l,mec}^{out} + \sigma_{l,th}^{out} \quad (2.36)$$

The Primary + Secondary stresses values are visible in the Table 7.

Table 2.5: Total stress components (Primary + Secondary) evaluated at the inner and outer boundaries of the AP1000 cladding wall, at the peak points.

Component / Direction	Inner Cladding Wall [MPa]	Outer Cladding Wall [MPa]
Hoop (tangential)	52.30	87.89
Radial	-27.25	-15.51
Longitudinal	0.40	47.72

2.6 ASME BPVC Section III verification

The structural integrity assessment of the fuel pin cladding is performed in accordance with the provisions of the ASME Boiler and Pressure Vessel Code (BPVC), Section III. As previously seen, the code imposes a clear distinction between primary and secondary stress categories, based on their differing potential to cause structural damage to the component.

- **Primary Stresses:** These are stresses generated directly by the action of external or internal mechanical loads, represented in this analysis by the fission gas internal pressure ($p_i = p_{i,max}$). To ensure adequate safety margins under design conditions, the ASME code prescribes that the primary equivalent stress must satisfy the following limit:

$$\sigma_{max,pri} \leq \sigma_a = S_m \quad (2.37)$$

The equivalent stress according to Tresca's maximum shear stress criterion is evaluated using the wall-averaged (membrane) mechanical stress components:

$$\sigma_{pri,max} = \max \left(|\sigma_{h,mec}^{avg} - \sigma_{r,mec}^{avg}|, |\sigma_{r,mec}^{avg} - \sigma_{l,mec}^{avg}|, |\sigma_{h,mec}^{avg} - \sigma_{l,mec}^{avg}| \right) \quad (2.38)$$

The allowable stress intensity limit is defined conservatively based on the material's yield strength (σ_y) and ultimate tensile strength (σ_u):

$$S_m = \min \left(\frac{2}{3}\sigma_y, \frac{1}{3}\sigma_u \right) \quad (2.39)$$

- **Primary + Secondary Stresses:** As these self-limiting loads combine both mechanical bending deviations and the severe radial temperature gradient across the cladding thickness, they are evaluated against an expanded design threshold. This higher limit accommodates localized plastic relaxation while ensuring protection against cyclic thermal failure modes.

The combined Primary + Secondary (Total) stress intensity must satisfy the following condition:

$$\sigma_{\max, \text{tot}} \leq \sigma_a = 3S_m \quad (2.40)$$

Since the thermal contribution varies drastically between boundaries, the total principal stresses are evaluated locally. The equivalent stress intensities are calculated using Tresca's criterion on both the inner cladding wall ($r = r_i$) and the outer cladding wall ($r = r_o$):

$$\sigma_{\text{tot}, \max}^{\text{in}} = \max(|\sigma_{h, \text{tot}}^{\text{in}} - \sigma_{r, \text{tot}}^{\text{in}}|, |\sigma_{r, \text{tot}}^{\text{in}} - \sigma_{l, \text{tot}}^{\text{in}}|, |\sigma_{h, \text{tot}}^{\text{in}} - \sigma_{l, \text{tot}}^{\text{in}}|) \quad (2.41)$$

$$\sigma_{\text{tot}, \max}^{\text{out}} = \max(|\sigma_{h, \text{tot}}^{\text{out}} - \sigma_{r, \text{tot}}^{\text{out}}|, |\sigma_{r, \text{tot}}^{\text{out}} - \sigma_{l, \text{tot}}^{\text{out}}|, |\sigma_{h, \text{tot}}^{\text{out}} - \sigma_{l, \text{tot}}^{\text{out}}|) \quad (2.42)$$

The structural verification results and their comparison with the allowable regulatory thresholds are summarized in Table 2.6.

Table 2.6: ASME Code Section III structural verification results for the AP1000 fuel pin cladding.

Component (TRESCA)	Inner Wall [MPa]	Outer Wall [MPa]	Limit [MPa]	Status
Primary (S_m)		92.07	137.67	PASSED
Primary + Secondary ($3S_m$)	79.55	103.40	413.00	PASSED

2.6.1 Results from the verification

In the simulation performed, **both structural checks successfully passed**, confirming that the cladding remains within the elastic regime and satisfies the safety margins under nominal operation.

However, to ensure robustness, the algorithm incorporates a conditional safety routine. If the primary membrane stress limit were to fail ($\sigma_{\max, \text{pri}} > S_m$) under the highly conservative bounding pressure $p_{i, \max}$ from section 5, the algorithm analytically inverts the Lamé-Tresca criterion at the inner radius. By setting the equivalent primary stress exactly equal to the allowable limit S_m , the maximum allowable internal pressure p_x is recalculated as:

$$p_x = \frac{S_m(r_o^2 - r_i^2) + 2p_o r_o^2}{2r_o^2} \quad (2.43)$$

Where p_o represents the external operating pressure of the refrigerant. Since the primary check passed in this design case, no pressure derating was necessary, and the reference pressure p_x for the subsequent gas plenum sizing was consistently set to the full bounding value:

$$p_x = p_{i, \max} = 27.25 \text{ MPa} \quad (272.52 \text{ bar}) \quad (2.44)$$

2.7 Gas plenum size

During reactor operations, high burn-up conditions prompt the accumulation of internal gases within the fuel rod cladding, creating a critical primary pressure load. This gas mixture originates from two distinct phenomena defined by the problem statement:

1. **Fission Product Generation:** The nuclear fissions yield volatile noble gases (Xe and Kr) that are progressively released from the UO_2 crystalline matrix into the free volume.
2. **Manufacturing Impurities:** Initial structural fabrication leaves trace chemical impurities trapped within the fuel pellets, namely nitrogen (N_2) and moisture (H_2O vapor).

In accordance with the design constraints, all volatile inventory is assumed to behave as an ideal gas and is restricted solely to the upper gas plenum volume, neglecting any additional accumulation within the pellet-cladding gap.

To find the absolute elemental yields, the average total mass of UO_2 per single fuel rod ($m_{\text{UO}_2} = 2.3530 \text{ kg}$) is derived from the assembly structural constraints. Since raw thermal burn-up is defined relative to the mass

of heavy metal atoms (t_U), the equivalent mass of pure Uranium per rod (m_U) must be calculated utilizing the fuel's specific molar mass fraction:

$$M_U = \epsilon \cdot M_{U^{235}} + (1 - \epsilon) \cdot M_{U^{238}} \quad (2.45)$$

$$m_U = m_{UO_2} \cdot \left(\frac{M_U}{M_{UO_2}} \right) = 2.0740 \text{ kg} = 2074 \text{ g} \quad (2.46)$$

Where $\epsilon = 0.03$ represents the baseline U^{235} enrichment fraction.

The total thermal energy produced by an individual fuel pin throughout its lifetime (E_{tot}) depends on the burn-up ($BU = 60000 \text{ MWD/t}_U$):

$$E_{\text{tot}} = BU \cdot \left(\frac{m_U}{1000} \right) \cdot (24 \cdot 3600 \cdot 10^6) \quad [\text{J}] \quad (2.47)$$

Given an average energy per fission event of $E_{\text{fiss}} = 200 \text{ MeV}$, the number of fissions (N_{fiss}) inside the rod is evaluated as:

$$N_{\text{fiss}} = \frac{E_{\text{tot}}}{E_{\text{fiss}} \cdot 1.60218 \times 10^{-13}} = 3.355 \times 10^{23} \text{ fissions} \quad (2.48)$$

Applying the fission yield for noble gases ($Y_{\text{Xe+Kr}} = 28\%$) and the corresponding fractional release rate out of the solid fuel pellet ($f_R = 40\%$), the net moles of released fission gas ($n_{\text{fiss.gas}}$) is quantified via Avogadro's constant ($N_A = 6.02214 \times 10^{23} \text{ atoms/mol}$):

$$n_{\text{fiss.gas}} = \frac{N_{\text{fiss}} \cdot Y_{\text{Xe+Kr}} \cdot f_R}{N_A} = 0.062402 \text{ mol} \quad (2.49)$$

Initial concentrations of trapped fabrication impurities are specified in parts per million by weight ($\text{ppm}_w = \text{mg/kg}$), corresponding to $\omega_{N_2} = 25 \times 10^{-6}$ e $\omega_{H_2O} = 75 \times 10^{-6}$. The molar concentrations of trapped nitrogen gas and moisture vapor are governed by their respective molecular weights ($M_{N_2} = 28.02 \times 10^{-3} \text{ kg/mol}$ e $M_{H_2O} = 18.016 \times 10^{-3} \text{ kg/mol}$):

$$n_{N_2} = \frac{m_{UO_2} \cdot \omega_{N_2}}{M_{N_2}} = \frac{2.3155 \text{ kg} \cdot (25 \times 10^{-6})}{28.02 \times 10^{-3} \text{ kg/mol}} = 0.002099 \text{ mol} \quad (2.50)$$

$$n_{H_2O} = \frac{m_{UO_2} \cdot \omega_{H_2O}}{M_{H_2O}} = \frac{2.3155 \text{ kg} \cdot (75 \times 10^{-6})}{18.016 \times 10^{-3} \text{ kg/mol}} = 0.009795 \text{ mol} \quad (2.51)$$

The total gas inventory to be contained by the plenum (n_{tot}) is the summation of all species:

$$n_{\text{tot}} = n_{\text{fiss.gas}} + n_{N_2} + n_{H_2O} = 0.074297 \text{ mol} \quad (2.52)$$

The internal pressure of the bounding design (p_{plenum}) is established equal to the maximum internal pressure (p_x), calculated during the mechanical verification of the preliminary sealing.

$$p_{\text{plenum}} = p_x \quad [\text{MPa}] \quad (2.53)$$

According to the thermal design parameters, the gas plenum temperature is equal to the inner cladding temperature at the same (axial) position ($T_{\text{plenum}} = T_{ci, \text{top}} = 346.10^\circ\text{C} = 619.25 \text{ K}$).

Utilizing the ideal gas law, the minimum required free internal volume of the gas plenum (V_{plenum}) is derived as follows:

$$V_{\text{plenum}} = \frac{n_{\text{tot}} \cdot R \cdot T_{\text{plenum}}}{p_{\text{plenum}}} = 14.04 \text{ cm}^3 \quad (1.404 \times 10^{-5} \text{ m}^3) \quad (2.54)$$

Where $R = 8.3144 \text{ J/(mol} \cdot \text{K)}$ represents the universal ideal gas constant. Given the inner radius of the Zircaloy cladding (r_i), the inner cross-sectional flow area is $A_{\text{int}} = \pi r_i^2$. The active minimum physical height required for gas containment (H_{plenum}) yields:

$$H_{\text{plenum}} = \frac{V_{\text{plenum}}}{A_{\text{int}}} = \frac{V_{\text{plenum}}}{\pi r_i^2} = 25.59 \text{ cm} \quad (0.2559 \text{ m}) \quad (2.55)$$

To arrive at the final engineering design specification (H_{final}) for manufacturing, an additional constant overhead length ($H_{\text{spring}} = 15.0 \text{ cm}$) must be allocated to accommodate the structural hold-down plenum spring:

$$H_{\text{final}} = H_{\text{plenum}} + H_{\text{spring}} = 25.59 + 15.0 = 40.59 \text{ cm} \quad (0.4059 \text{ m}) \quad (2.56)$$

2.8 Thermodynamic Validation of the Ideal Gas Assumption for H_2O Vapor

To rigorously assess the validity of treating the H_2O vapor as an ideal gas within the gas plenum, a dedicated thermodynamic check must be performed. The validation strategy relies on quantifying the ideal partial pressure of the moisture inventory within the calculated plenum volume (V_{plenum}) and comparing it against the saturation pressure of pure water (p_{sat, H_2O}) evaluated at the local plenum temperature (T_{plenum}). The ideal partial pressure of water vapor ($p_{H_2O, \text{id}}$) is defined via Dalton's Law:

$$p_{H_2O, \text{id}} = \frac{n_{H_2O} \cdot R \cdot T_{\text{plenum}}}{V_{\text{plenum}}} = 35.9288 \text{ bar} = 3592876 \text{ Pa} \quad (2.57)$$

To determine the true thermodynamic state of the fluid, the saturation pressure is dynamically queried using high-fidelity Helmholtz energy formulations implemented in the **CoolProp** library:

$$p_{\text{sat}, H_2O} = \text{PropsSI}('P', 'T', T_{\text{plenum}}, 'Q', 0, 'Water') \quad [\text{Pa}] \quad (2.58)$$

Where $Q = 0$ specifies the saturated liquid state boundary for the given temperature. CoolProp returns the pressure in Pascal, but converted to bar it is equal to 157.54 bar.

Furthermore, to mathematically justify the ideal gas hypothesis ($pV = nRT$), the deviation of the real gas behavior from ideal conditions must be quantified through the **compressibility factor** Z , defined as:

$$Z = \frac{p_{H_2O, \text{id}} \cdot V_{\text{real}}}{n_{H_2O} \cdot R \cdot T_{\text{plenum}}} \quad (2.59)$$

For an ideal gas, $Z \equiv 1$. In the algorithm, this parameter is dynamically evaluated at the local partial pressure and temperature using **CoolProp**:

$$Z_{\text{water}} = \text{PropsSI}('Z', 'T', T_{\text{plenum}}, 'P', p_{H_2O, \text{id}}, 'Water') \quad (2.60)$$

An engineering acceptance tolerance of 10% ($|1.0 - Z_{\text{water}}| \leq 0.10$) is established as the threshold for validity. Considering the critical water temperature almost equal to 373.95 °C, the engineering acceptance criteria for hypothesis testing are formulated as follows:

- **Subcritical Condensation Check** ($T_{\text{plenum}} \leq 373.95^\circ\text{C}$): If $p_{H_2O, \text{id}} < p_{\text{sat}, H_2O}$ and the compressibility factor satisfies $|1.0 - Z_{\text{water}}| \leq 0.10$, water remains entirely in the superheated vapor phase and behaves closely to an ideal gas. This confirms that the ideal gas assumption is acceptable for design purposes. Conversely, if either condition is violated, the ideal gas law is rendered invalid.
- **Supercritical Phase State Check** ($T_{\text{plenum}} > 373.95^\circ\text{C}$): If the plenum temperature exceeds the thermodynamic critical temperature of water, liquid-vapor phase coexistence ceases to exist. In this supercritical regime, distinct physical condensation is impossible regardless of pressure, which justifies the use of gas-like state equations as an acceptable engineering approximation.

Given the current thermal-hydraulic profile where $T_{\text{plenum}} = 346.10^\circ\text{C}$ (619.25 K), the system operates within the subcritical domain, close to the critical point. The simulation yields a compressibility factor of $Z_{\text{water}} = 0.931$. Since the calculated ideal partial pressure (35.93 bar) remains well below the corresponding saturation pressure threshold (157.54 bar) and the deviation of Z_{water} from unity is only 6.9% (well within the 10% tolerance), the check successfully passes. The H_2O inventory stays completely in the vapor state and satisfies the ideal gas modeling assumptions, validating the structural plenum sizing calculations..

Chapter 3

Final plots

3.1 Methodological transition: from localized peak properties to axial vector mapping

Initially, the core thermo-mechanical stress evaluation in the previous steps was carried out using a conservative, localized approach. Under this formulation, the mechanical and thermal properties of the Zircaloy-4 cladding, specifically Young's modulus (E), Poisson's ratio (ν), and the linear thermal expansion coefficient (α), were evaluated exclusively at the absolute peak temperature nodes ($T_{\text{peak,in}}$ and $T_{\text{peak,out}}$) where the radial temperature difference (ΔT_{cl}) reaches its maximum value. This localized calculation was instrumental in establishing a bounding upper limit for the structural integrity verification.

However, to capture the continuous structural behavior across the entire active core height (z), a refined axial vector mapping was implemented for the comprehensive stress plotting. Since the cladding temperature experiences a variation of over 100°C between the core inlet and outlet, treating material properties as constants would lead to a non-physical overestimation or underestimation of stresses along the channel.

Therefore, in this final stage, the Zircaloy-4 properties ($E(z)$, $\nu(z)$, and $\alpha(z)$) were dynamically recalculated as continuous vectors, tracked node-by-node according to the local inner and outer temperature profiles ($T_{\text{ci_vec}}$ and $T_{\text{co_vec}}$). This distributed approach allows for an accurate physical representation of the pure thermal profiles and total combined stresses along the active fuel rod geometry.

3.2 Hoop stress distribution (inner vs outer cladding wall)

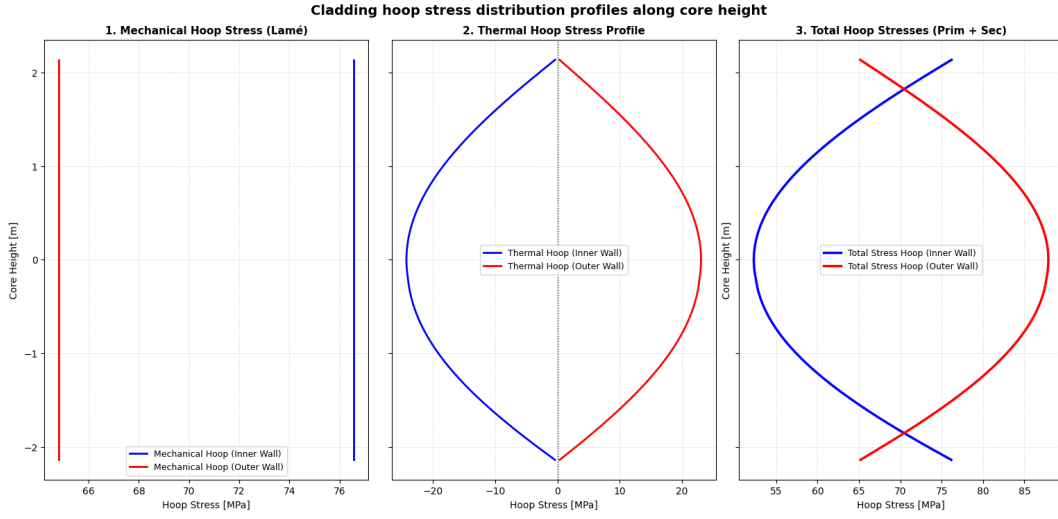


Figure 3.1: Thermo-mechanical hoop stress profiles along the active core height.

The plotted profiles show that while the pure mechanical hoop stress remains uniform along the core height due to both the internal bounding pressure ($p_{i,\max} = 27.25$ MPa) and the external operating coolant pressure ($p_o = 15.51$ MPa), the thermal hoop stress follows a chopped-cosine shape driven by the core power distribution. On the hotter inner wall, the thermal stress is compressive, acting as a beneficial counter-load that reduces the total combined stress near the mid-plane. Conversely, on the colder outer wall, the thermal stress is tensile and adds directly to the mechanical load, peaking at the core center.

3.3 Longitudinal stress distribution (inner vs outer cladding wall)

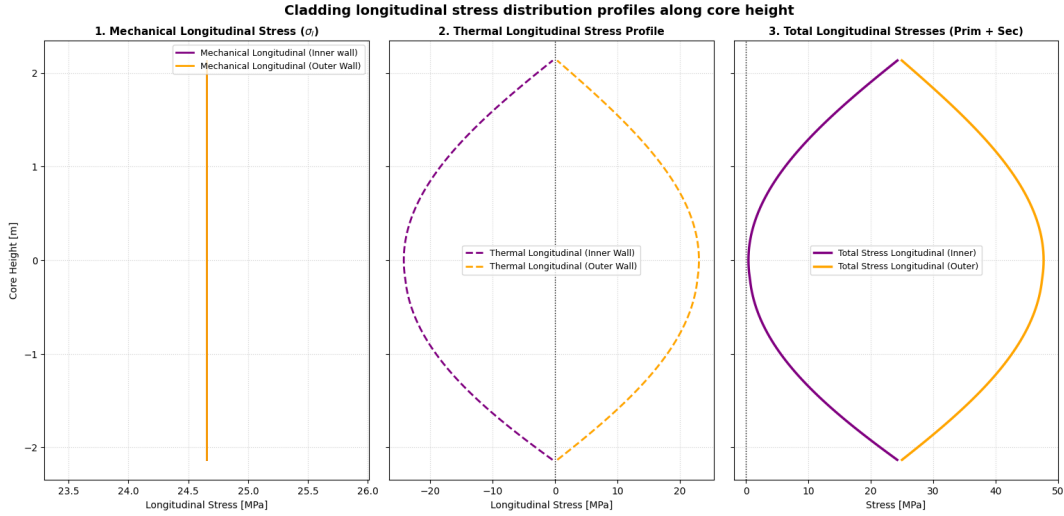


Figure 3.2: Thermo-mechanical longitudinal stress profiles along the active core height.

The longitudinal stress profiles display a physical behavior consistent with thick-walled closed-cylinder theory. In Subplot 1, the pure mechanical longitudinal stress is uniform and perfectly superimposed at 24.65 MPa for

both walls. The thermal longitudinal stress (Subplot 2) follows a chopped-cosine profile, inducing compression (-24.26 MPa) on the hotter inner wall and traction (23.07 MPa) on the colder outer wall. In the combined profile (Subplot 3), the thermal compression cancels out the mechanical tension at the mid-plane along the inner radius, reducing total stress to 0.40 MPa. Conversely, on the outer wall, they superimpose constructively to create a peak tensile stress of 47.72 MPa.

3.4 Radial stress distribution (inner vs outer cladding wall)

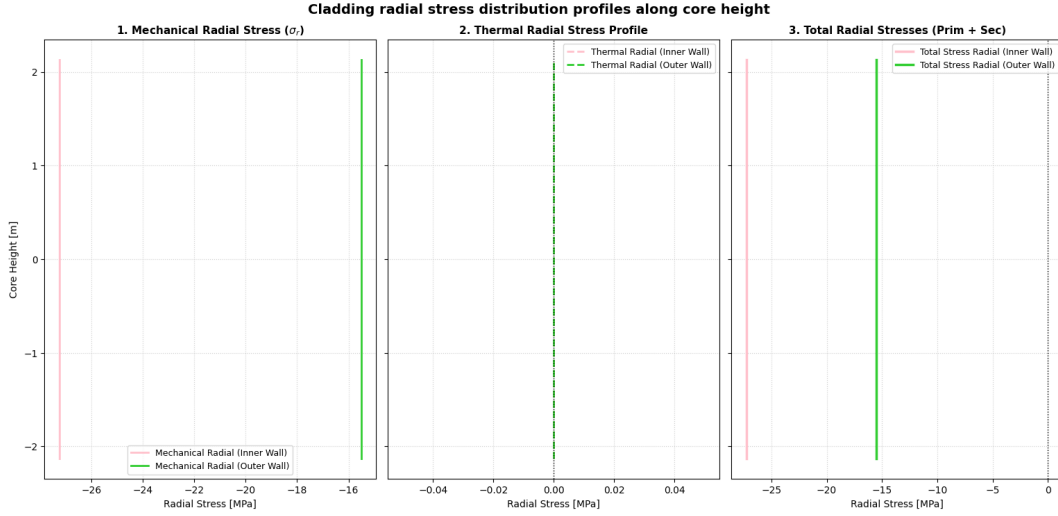


Figure 3.3: Thermo-mechanical radial stress profiles and ASME Section III verification along the active core height.

From the Figure 3 is possible to notice that the mechanical radial stress profiles are constant along the axial direction. The thermal radial stress is zero on both walls. As a direct consequence, the total radial stress is identical to the mechanical one.

3.5 Compare between ASME Section III with real thermo-mechanical stresses

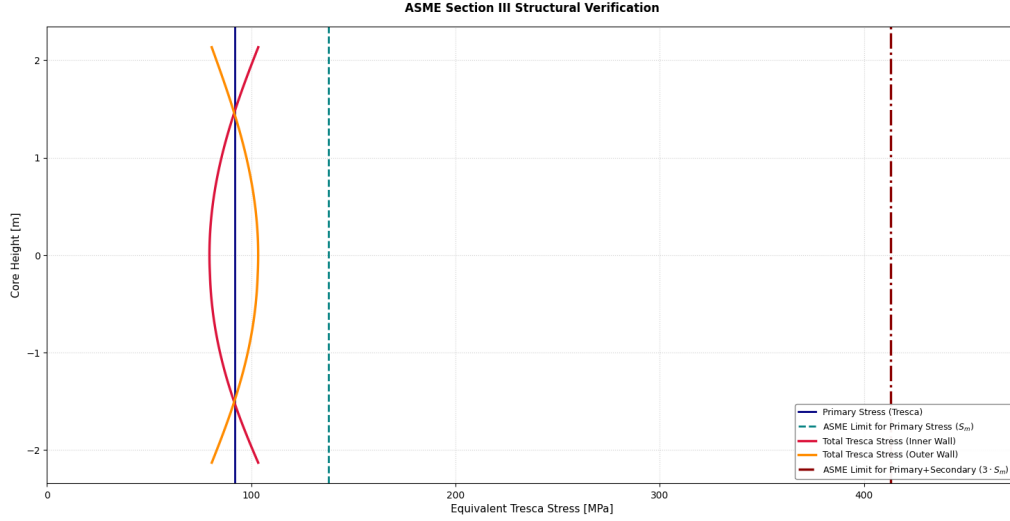


Figure 3.4: ASME Section III verification profiles

The ASME verification plot consolidates the result of the entire thermo-mechanical analysis. The primary membrane stress intensity (92.07 MPa, blue line) is uniform and remains safely below the primary design limit ($S_m = 137.67$ MPa, teal dashed line). When considering total combined loading (primary + secondary), the Tresca equivalent stress profiles for both the inner wall (red curve) and outer wall (orange curve) vary along the active core height due to the axial temperature profile, reaching a maximum peak of 103.40 MPa on the outer surface. Both total stress curves stay significantly below the strict secondary stress intensity threshold ($3S_m = 413.00$ MPa, dark red dash-dotted line), structurally demonstrating that the AP1000 fuel pin preserves extensive margins against plastic deformation during steady-state nominal operations

Chapter 4

Conclusions

In conclusion, a comprehensive thermo-mechanical verification of an AP1000 Zircaloy-4 fuel rod cladding has been successfully performed under nominal steady-state operating conditions. The structural and thermal response of the component was evaluated step-by-step through analytical model implementation, culminating in the engineering sizing and thermodynamic validation of the gas plenum.

The preliminary buckling evaluation demonstrated that the cladding possesses a substantial structural margin against external collapse during depressurization or early-life operation. The computed critical buckling pressure ($p_{cr} = 459.70$ bar) exceeds the reactor system operating pressure ($P = 155.13$ bar) by a safety margin of 304.57 bar, ensuring that the sheath is entirely safe against elastic instability.

The core mechanical analysis revealed that under a geometric ratio of $t/r_{avg} = 0.13$, the cladding must be rigorously treated as a thick-walled cylinder via Lamé’s equations rather than a thin tube approximation. When applying the internal pressure of the conservative design (27.25 MPa), which assumes complete depressurization of the reactor, the Tresca stress resulting from the primary membrane reached 92.07 MPa at the inner wall. This value satisfies the strict primary design limit established by the ASME BPVC Section III standard ($S_m = 137.67$ MPa).

The secondary stresses induced by the radial thermal gradient across the cladding wall (-24.26 MPa at the inner wall and 23.07 MPa at the outer wall) provided a characteristic stress redistribution when combined with the mechanical profile. The complete primary plus secondary stress intensity reaches its peak on the outer surface at 103.40 MPa (and 79.56 MPa at the inner radius), easily satisfying the ASME Service Level A limit ($3S_m = 413.00$ MPa) and leaving a wide safety margin for steady-state operations.

Finally, the manufacturing constraints for gas containment were successfully defined. To accommodate the cumulative inventory of noble fission gases (Xe and Kr released at a high burn-up of 60000 MWD/ t_U) alongside initial trapped fabrication impurities (N_2 and H_2O), a minimum free plenum volume of 14.04 cm³ is required. This volume translates into a minimum active physical height of 25.59 cm, which perfectly aligns with standard industrial PWR design references (~ 20 cm). Including the 15.0 cm design allowance for the top hold-down spring, the total engineering specification for the gas plenum height is fixed at 40.59 cm, which is fully compatible with the overall length of an AP1000 fuel assembly without penalizing the active fuel length.

The engineering robustness of this sizing was further validated by the thermodynamic check performed on the moisture impurity. At the local operating temperature of 346.10°C, the ideal partial pressure of water vapor (35.93 bar) remains significantly below the saturation threshold (157.54 bar). Furthermore, the compressibility factor queried via CoolProp ($Z_{water} = 0.9310$) shows a maximum deviation from ideal behavior of only 6.90%, well within the 10% design tolerance. The water inventory is thus guaranteed to remain entirely within the superheated vapor phase without any risk of subcritical condensation, validating the use of the ideal gas state equation as a highly accurate and physically sound approximation for the design verification of the AP1000 fuel pin.

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